

as PNG, PDF, etc, require *Cairo* Julia bindings, which is a 2D graphics library (<https://github.com/JuliaGraphics/Cairo.jl>):

```
samplePlot = Gadfly.plot(layer(x=rand(10), y=rand(10), Geom.  
point, order=1),  
    layer(x=rand(10), y=rand(10), Geom.Line, order=2))  
draw(SVG("myplot.svg", 4inch, 3inch), samplePlot)  
draw(PDF("myplot.pdf", 4inch, 3inch), samplePlot)
```

Themes

Gadfly has various options to customise the plot with themes. A sample code is shown below and its output is given in Figure 8. In this plot, the panel background colour is *gray* and the *default_color* is set as *orange*.

```
using Gadfly  
dark_panel = Theme(  
    panel_fill=Colorant{Gray},  
    default_color=Colorant{Orange}  
)  
Gadfly.plot(x=rand(10), y=rand(10), dark_panel)
```

The plots can be customised with various parameters. Some of them are listed below:

- *line_width*
- *panel_fill*
- *panel_opacity*
- *grid_color*
- *discrete_color_scheme*
- *continuous_color_scheme*

Geometries

The core drawing of the plot is made by geometries, and those which are available are represented using 'Geom'. Gadfly provides more than twenty styles. Some of the Geom types are listed below:

- *Geom.point*
- *Geom.line*
- *Geom.polygon*
- *Geom.boxplot*
- *Geom.ribbon*
- *Geom.violin*

The complete list of all Geoms is available at <http://gadflyjl.org/stable/man/geometries.html>.

For example, a Boxplot with RDataSets can be easily built as shown below:

```
using Gadfly  
using RDataSets  
Gadfly.plot(dataset("lattice", "singer"), x="VoicePart",  
y="Height", Geom.boxplot)
```

To summarise, the features provided by Gadfly are

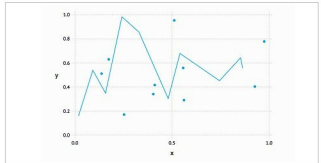


Figure 7: Gadfly – stacking and layering

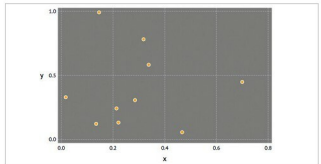


Figure 8: Gadfly-customising plots with themes

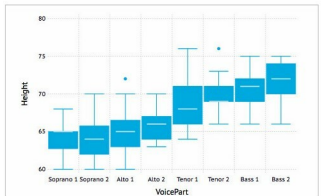


Figure 9: Boxplot

really useful in making publication-quality plots. At the same time, making these plots is fairly simple. This excellent combination of features makes Gadfly a great choice with respect to visualisation of data. **END**

References

- [1] <http://gadflyjl.org/stable/index.html>
- [2] <https://juliabox.com/>
- [3] <https://github.com/GiovineItalia/Gadfly.jl>

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